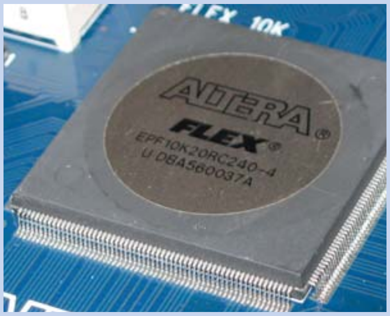
Been There, Done That?

The idea of a reconfigurable chip that can take on many different roles isn't new, however. It's been around in the form of the Field Programmable Gate Array (FPGA), a semiconductor device built from logic gates, with reconfigurable interconnects between them. These were the original "stem cells" of the electronic world—out of the foundry in no particular form, and then configured depending on their intended use.



The Altera Flex FPGA is one of the more commercially popular FPGA chips

The first FPGAs were configurable only once, then came the era of the reprogrammable FPGA—which allowed you to rewire the interconnects as and when you needed to. Soon enough, the FPGA was proposed as a replacement for embedded microcontrollers, culminating in the manufacture of fully-functional CPUs. Xilinx's MicroBlaze and Altera's Nios II are both "soft CPU cores" based on FPGAs (called "soft" because you could reconfigure them if you wanted to), but they found little mainstream use as they were remarkably slower than their specifically-built counterparts.

You'd expect that a chaotic chip would be based on the FPGA, but you would be quite wrong. The trouble with the FPGA is that it takes a few milliseconds to reconfigure the interconnects for a new operation—highly unacceptable. A chaotic chip, however, will be able to change its role in just one clock cycle, making it (quite literally) a million times faster.

Pandemonium Ahead

In March 2005, Indian scientists K Murali and Sudeshna Sinha, along with William Ditto, published their first experiments with building a real-life reconfigurable cell, and implemented a binary half-adder with it. They've also had some success with building a chaotic element using a leech neuron, though it will still be time before we see nerve cells in our PCs.

If all goes well (and even the creators admit that's a big if), we'll probably see the first commercially viable chaotic chips before the decade runs out. But then, they said similar things about quantum computers, and we all know what's become of *that* story. The truth is, there's a lot that might hinder chaos computing—from design issues to inherent electrical flaws that can't be removed, to the possibility that programmers will find it too difficult and throw the concept to the dogs. Still, the idea is a lot more appealing than building new chips from the ground up, and definitely shows more promise of becoming a reality sooner than quantum or DNA computing. When it does get off the ground, well, your imagination will be the only one limiting the possibilities–smaller, smarter cell phones with days worth of talk-time, faster CPUs that spend their spare time as graphics and sound cards, PCs that can heal themselves—take your pick!

And you thought chaos was a bad thing. ☒

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